Note:

i. All questions are compulsory.
ii. Figures to the right indicate full marks.
iii. Graph of L.P.P. should be drawn on graph paper only.
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SECTION – I

Q.1. (A) Select and write the appropriate answer from the given alternatives in each of the following sub-questions: **(6) [12]**

i. If $A = \begin{bmatrix} 2 & -3 \\ 4 & 1 \end{bmatrix}$, then adjoint of matrix A is _____.

(A) $\begin{bmatrix} 1 & 3 \\ -4 & 2 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & -3 \\ -4 & 2 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 3 \\ 4 & -2 \end{bmatrix}$ (D) $\begin{bmatrix} -1 & -3 \\ -4 & 2 \end{bmatrix}$

ii. The principal solutions of $\sec x = \frac{2}{\sqrt{3}}$ are

(A) $\frac{\pi}{3}, \frac{11\pi}{6}$ (B) $\frac{\pi}{6}, \frac{11\pi}{6}$

(C) $\frac{\pi}{4}, \frac{11\pi}{4}$ (D) $\frac{\pi}{6}, \frac{11\pi}{4}$

iii. The measure of acute angle between the lines whose direction ratios are 3, 2, 6 and –2, 1, 2 is _____.

(A) $\cos^{-1}\left(\frac{1}{7}\right)$ (B) $\cos^{-1}\left(\frac{8}{15}\right)$

(C) $\cos^{-1}\left(\frac{1}{3}\right)$ (D) $\cos^{-1}\left(\frac{8}{21}\right)$

(B) Attempt any THREE of the following: **(6)**

i. Write the negations of the following statements:
   a. All students of this college live in the hostel.
   b. 6 is an even number or 36 is a perfect square.

ii. If a line makes angles α, β, γ with the co-ordinate axes, prove that $\cos 2\alpha + \cos 2\beta + \cos 2\gamma + 1 = 0$.

iii. Find the distance of the point (1, 2, –1) from the plane $x - 2y + 4z - 10 = 0$.

iv. Find the vector equation of the line which passes through the point with position vector $4\hat{i} - \hat{j} + 2\hat{k}$ and is in the direction of $-2\hat{i} + \hat{j} + \hat{k}$.

v. If $\bar{a} = 3\hat{i} - 2\hat{j} + 7\hat{k}$, $\bar{b} = 5\hat{i} + \hat{j} - 2\hat{k}$ and $\bar{c} = \hat{i} + \hat{j} - \hat{k}$, then find $\bar{a} \cdot (\bar{b} \times \bar{c})$.

Q.2. (A) Attempt any TWO of the following: **(6)[14]**

i. Using vector method prove that the medians of a triangle are concurrent.

ii. Using the truth table, prove the following logical equivalence:

$p \leftrightarrow q \equiv (p \wedge q) \vee (\sim p \wedge \sim q)$.

iii. If the origin is the centroid of the triangle whose vertices are A(2, p, −3), B(q, −2, 5) and R(−5, 1, r), then find the values of p, q, r.

(B) Attempt any TWO of the following: **(8)**

i. Show that a homogeneous equation of degree two in x and y, i.e. $ax^2 + 2hxy + by^2 = 0$ represents a pair of lines passing through the origin if $h^2 - ab \geq 0$.

ii. In ΔABC, prove that $\tan\left(\frac{C-A}{2}\right) = \left(\frac{c-a}{c+a}\right)\cot\frac{B}{2}$

iii. Find the inverse of the matrix, $A = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$ using elementary row transformations.

Q.3. (A) Attempt any TWO of the following: **(6)[14]**

i. Find the joint equation of the pair of lines passing through the origin, which are perpendicular to the lines represented by $5x^2 + 2xy - 3y^2 = 0$.

ii. Find the angle between the lines $\frac{x-1}{4} = \frac{y-3}{1} = \frac{z}{8}$ and $\frac{x-2}{2} = \frac{y+1}{2} = \frac{z-4}{1}$

iii. Write converse, inverse and contrapositive of the following conditional statement:
If an angle is a right angle then its measure is 90°.

(B) Attempt any TWO of the following: **(8)**

i. Prove that: $\sin^{-1}\left(\frac{3}{5}\right) + \cos^{-1}\left(\frac{12}{13}\right) = \sin^{-1}\left(\frac{56}{65}\right)$

ii. Find the vector equation of the plane passing through the points A(1, 0, 1), B(1, −1, 1) and C(4, −3, 2).

iii. Minimize $Z = 7x + y$ subject to
$5x + y \geq 5, x + y \geq 3, x \geq 0, y \geq 0$

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SECTION – II

Q.4. (A) Select and write the appropriate answer from the given alternatives in each of the following sub-questions: **(6)[12]**

i. Let the p.m.f. of a random variable X be –

$P(x) = \frac{3-x}{10}$ for $x = -1, 0, 1, 2$

$= 0$ otherwise

Then E(X) is _____.

(A) 1 (B) 2

(C) 0 (D) −1

ii. If $\int_0^k \frac{1}{2+8x^2}\,dx = \frac{\pi}{16}$, then the value of k is _____.

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$

(C) $\frac{1}{4}$ (D) $\frac{1}{5}$

iii. Integrating factor of the linear differential equation $x\frac{dy}{dx} + 2y = x^2 \log x$ is _____.

(A) $\frac{1}{x^2}$ (B) $\frac{1}{x}$

(C) x (D) x^2

(B) Attempt any THREE of the following: **(6)**

i. Evaluate: $\int e^x \left[\frac{\cos x - \sin x}{\sin^2 x}\right] dx$

ii. If $y = \tan^2(\log x^3)$, find $\frac{dy}{dx}$.

iii. Find the area of ellipse $\frac{x^2}{1} + \frac{y^2}{4} = 1$.

iv. Obtain the differential equation by eliminating the arbitrary constants from the following equation: $y = c_1e^{2x} + c_2e^{-2x}$

v. Given $X \sim B(n, p)$
If n = 10 and p = 0.4, find E(X) and Var. (X).

Q.5. (A) Attempt any TWO of the following: **(6)[14]**

i. Evaluate: $\int \frac{1}{3 + 2\sin x + \cos x}\, dx$

ii. If $x = a\cos^3 t$, $y = a\sin^3 t$,

show that $\frac{dy}{dx} = -\left(\frac{y}{x}\right)^{\frac{1}{3}}$

iii. Examine the continuity of the function:

$f(x) = \frac{\log 100 + \log(0.01 + x)}{3x}$, for $x \neq 0$

$= \frac{100}{3}$ for $x = 0$; at $x = 0$

(B) Attempt any TWO of the following: **(8)**

i. Find the maximum and minimum value of the function:
$f(x) = 2x^3 - 21x^2 + 36x - 20$.

ii. Prove that: $\int \frac{1}{a^2 - x^2} dx = \frac{1}{2a}\log\left|\frac{a+x}{a-x}\right| + c$

iii. Show that: $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$, if $f(x)$ is an even function.

$= 0$, if $f(x)$ is an odd function.

Q.6. (A) Attempt any TWO of the following: **(6)[14]**

i. If $f(x) = \frac{x^2 - 9}{x - 3} + \alpha$, for $x > 3$

$= 5$, for $x = 3$

$= 2x^2 + 3x + \beta$, for $x < 3$

is continuous at $x = 3$, find α and β.

ii. Find $\frac{dy}{dx}$ if $y = \tan^{-1}\left(\frac{5x+1}{3 - x - 6x^2}\right)$

iii. A fair coin is tossed 9 times. Find the probability that it shows head exactly 5 times.

(B) Attempt any TWO of the following: **(8)**

i. Verify Rolle's theorem for the following function:
$f(x) = x^2 - 4x + 10$ on $[0, 4]$

ii. Find the particular solution of the differential equation:

$y(1 + \log x)\frac{dx}{dy} - x\log x = 0$

when $y = e^2$ and $x = e$

iii. Find the variance and standard deviation of the random variable X whose probability distribution is given below:

x	0	1	2	3
$P(X = x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$